

Ellann Cohen

ellann.cohen@gmail.com • (503) 505-3695 • <https://www.linkedin.com/in/ellann/>

Career Objective

Pursing leadership role in data science utilizing my background to advance the fields of thermal, mechanical, and quality/reliability engineering in the silicon industry

Industry Experience

Product Quality and Reliability Engineer, Product Quality and Reliability
Intel Corporation, Aug 2022 – Current

- Rapidly backfilled for the Extended Life Experiments owner role mid-experiments and product launch and got the last variants of the product to launch.
- Executed the next gen Extended Life Experiments and completed JMP analysis. Found a significant bug and led the resolution in a multi-functional task force. Work enabled 5-week pull-in for product launch.
- Owned thermal mechanical experiments (thermal cycling, bake, humidity) on future server product with a challenging new packaging technology, finding and resolving issues in a cross-organizational effort.

Thermal Analyst and Designer, Client Computing Group
Intel Corporation, January 2017 – August 2023

Hillsboro, OR

- Used linear regression to build a model that predicts the temperature on a thermal solution without the need for an additional thermal couple that was deployed with the Intel Cryo Cooler Thermal Solution
- Built neural network to predict performance of heat exchangers in the early design phase.
- Processed gigabytes of time series data - plotting, using edge detection to separate data into individual trials, finding trends, patterns, and insights into system and trial variation
- Coordinating and running TIM2 characterization efforts in-situ with teams across multiple geos
- Led the transformation of an outdated wiki to the current hub of training and best-known method (BKM) communication for a team that expanded from 10 to 27 employees over a short 1.5 year period. Now being adopted by sister teams.
- Spearheaded a comprehensive college level equivalent semester long course of content used to train new thermal analysts. Organized delivery and recording of this content in lecture format 2x a week for 4 months. Trained 25+ individuals with material and personally mentored 7.

Mechanical Design Engineer, Client Computing Group
Intel Corporation, January 2014 – January 2017

Hillsboro, OR

- Owned mechanical PCB integration for multiple boards/FPCs, ID interaction, and all IO and buttons (Plastic injection molding & MIM) for Chromebook Reference Design from start thru final assembly
- Early concept development & prototyping for new form factors as well as early-stage thermal testing

Rotation Engineering Program
Intel Corporation, January 2012 – January 2014

Hillsboro, OR – Albuquerque, NM – Santa Clara, CA

Mechanical Design/Development Engineer, Intel Media & PC Client Group & Intel Module Repair

- Designed multiple plastic injection molded enclosures to protect, hold, and label different software development vehicles and debug boards.
- Worked with vendors to design and manufacture stamped SMT EMI shields for multiple PCBs
- Reverse engineered and then redesigned a component of a factory tool to setup an internal repair loop and save capital expense

Human Factors Engineer/Interaction Designer, PC Client Group

- Designed and conducted multiple user studies surrounding user experience of touch on clamshells and keyboard comfort for reduced clamshell thickness
- Designed a new GUI for software project to shift the end user from OEMs to ‘techy’ consumers

Skills

Data skills: Python (Jupyter notebooks, pandas, matplotlib, numpy, flask), Matlab, SQL, R, tidyverse
Thermal/Mechanical: ProE, Solidworks, Fusion 360, Ansys Icepack, Flotherm / XT, Docea/iDTP

Physical Prototyping: 3D printing, Laser cutting, Arduino, Machinery (lathe, mill, drill press, band saw)

Publications External Conference Papers

Gaudin, G. and Cohen, E. and Cardenas, R. (July 2020) “Predicting Notebook Heat Exchanger Performance Using a Neural Network Approach.” Paper presented at ITHERM 2020 Virtual.

Internal to Intel

Cohen, E., & Cardenas, R. (2020). “Predicting Notebook Heat Exchanger Performance Using a Neural Network Approach” [Presentation]. Intel Mechanical and Thermal Engineering Summit 2020, Virtual.

Cohen, E., & Rallabandi, A. (2020). “Virtual Heat Sink Temperatures for Desktop Systems through CPU Telemetry.” [Poster]. Intel Mechanical and Thermal Engineering Summit 2020, Virtual.

Rallabandi, A., MacDonald, M., Cohen, E., & Carver, J. (2020). “Intel Cryo Cooling Technology” [Poster]. Intel Mechanical and Thermal Engineering Summit 2020, Virtual.

Cohen, E. (2019). “Simplified tool for transient predictions of silicon temperature with nonuniform power.” [Presentation]. Intel Mechanical and Thermal Engineering Summit 2019, Hillsboro, OR.

Kothari, S., Cohen, E., Guo, X., Mani, D., & Sebot, J. (May 2020) “Hot-spot & Early Architecture Thermals (HEAT) tool to enable power-thermal co-optimization of Silicon microarchitecture.” [Paper]. Design & Test Technology Conference 2019, Portland, OR.

x4 Invention Disclosure Forms accepted

Hobbies

Pacific Giant Vegetables Growers Association: Vice President, Volunteer & Participant, 2017 - current

Mazamas, Mountaineering Club: Beginning Mountaineering Teaching Assistant, 2018 - 2019

Bike & Build: Summer 2011: Completed cross-country cycling trip, helped with affordable housing

MIT Varsity Sailing Team: Skipper and crew, 2004 – 2008: *MVP for 2006-7 and 2007-8 Seasons*

Education

Massachusetts Institute of Technology (MIT)

M.S. in Mechanical Engineering, **GPA: 5.0/5.0**

B.S. in Mechanical Engineering, **GPA: 4.9/5.0**

Willamette University

M.S. in Data Science, **GPA: 4.0/4.0**